Stochastic Optimization for Macroeconomic Forecasting

A Sample Average Approximation Approach with Application to Serbian GDP

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Abstract

We develop a stochastic optimization framework for macroeconomic forecasting under data uncertainty. Classical forecasting models typically treat indicator data as perfectly observed, ignoring measurement error, revisions, and sampling uncertainty inherent in official statistics. We formulate the weight-selection problem as a stochastic program and study the corresponding Sample Average Approximation (SAA) estimator. Under standard regularity conditions, we establish almost sure consistency of the SAA estimator. For quadratic loss, we show that the expected objective takes a ridge-type regularized form induced by the noise covariance matrix. Empirically, we apply the framework to Serbian quarterly GDP forecasting and show that a stochastic lasso specification achieves statistical parity with standard benchmark methods such as Lasso, Ridge, and OLS, while clearly outperforming a naive random-walk baseline. The paper is positioned as a stochastic continuation of the Serbian composite-index literature developed by Vukmirović and coauthors.

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1 Introduction

Macroeconomic forecasting relies on combining multiple indicator variables into predictive models. A fundamental challenge is that economic data are measured with error: preliminary GDP estimates are routinely revised, survey indicators contain sampling noise, and financial variables exhibit measurement uncertainty. Despite this, standard approaches often treat the data as perfectly observed.

Let $y_t \in \mathbb{R}$ denote the target variable (e.g., GDP growth) at time t, and let $\mathbf{x}_t \in \mathbb{R}^p$ be a vector of p predictor variables. The classical forecasting problem seeks weights $\mathbf{w} \in \mathbb{R}^p$ minimizing the empirical loss:

$$\min_{\mathbf{w} \in \mathcal{W}} \frac{1}{T} \sum_{t=1}^{T} \ell(y_t, \mathbf{x}_t^\top \mathbf{w}), \tag{1}$$

where $\mathcal{W} \subseteq \mathbb{R}^p$ is a constraint set and $\ell : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_+$ is a loss function.

Problem (1) implicitly assumes that $\mathbf{x}_t$ is observed without error. In practice, we observe

$$\tilde{\mathbf{x}}_t = \mathbf{x}_t + \boldsymbol{\xi}_t,$$

where $\boldsymbol{\xi}_t$ represents measurement error. This paper develops a stochastic optimization framework that explicitly accounts for this uncertainty.

1.1 Relation to the Serbian composite-index literature

This paper should be read as a continuation of the Serbian literature on composite indicators of economic activity. Vukmirovic et al. (2009) developed the Serbian Composite Index as an instrument for measuring short-term movements in economic activity. Vukmirovic and Ciric (2010) then used the Serbian Composite Index for flash estimates of quarterly GDP. Later work extended the methodology toward a monthly GDP indicator through revised weighting and benchmarking procedures (Vukmirovic et al., 2013), while Vukmirovic et al. (2015) used the resulting monthly activity indicator for dating Serbian business cycles.

The present paper differs from that literature in one central respect. Earlier Serbian composite-index work is fundamentally deterministic: once the indicators, transformations, and weights are chosen, the index is computed as if the predictor information were observed without error. By contrast, this paper asks how the estimation problem changes when indicator variables are explicitly treated as noisy measurements. In that sense, the contribution here is not to replace the composite-index tradition, but to extend it by embedding forecasting and indicator weighting into a stochastic optimization framework.

1.2 Contributions

1. We formulate indicator-weight selection as a stochastic program under data uncertainty.
2. We establish conditions under which the Sample Average Approximation estimator is consistent.
3. For quadratic loss, we show that the stochastic objective induces a ridge-type regularization term driven by the noise covariance matrix.
4. We demonstrate the methodology on Serbian GDP forecasting using real data and position the exercise relative to the earlier Serbian composite-index literature.

2 Problem Formulation

2.1 The stochastic forecasting problem

We model the observed predictor as a noisy realization of the true predictor:

$$\tilde{\mathbf{x}}_t = \mathbf{x}_t + \boldsymbol{\xi}_t, \tag{2}$$

where $\boldsymbol{\xi}_t \in \mathbb{R}^p$ is a random measurement error.

Definition 2.1 (Stochastic loss function)**.** *Given a weight vector $\mathbf{w} \in \mathcal{W}$ and a loss function ℓ, define the stochastic loss by*

$$L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t) = \ell\big(y_t, (\mathbf{x}_t + \boldsymbol{\xi}_t)^\top \mathbf{w}\big). \tag{3}$$

Definition 2.2 (Expected loss)**.** *The expected loss under measurement uncertainty is*

$$f(\mathbf{w}) = \mathbb{E}_{\boldsymbol{\xi}} \left[\frac{1}{T} \sum_{t=1}^{T} L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t) \right], \tag{4}$$

where the expectation is taken over the error distribution.

Definition 2.3 (Stochastic optimization problem)**.** *The stochastic weight-selection problem is*

$$\mathbf{w}^* = \operatorname*{argmin}_{\mathbf{w} \in \mathcal{W}} f(\mathbf{w}). \tag{5}$$

2.2 Constraint set specifications

We consider two constraint sets relevant for macroeconomic applications.

Simplex constraints (composite index):

$$\mathcal{W}_{\text{simplex}} = \left\{ \mathbf{w} \in \mathbb{R}^p : w_j \geq 0 \ \forall j, \ \sum_{j=1}^{p} w_j = 1 \right\}. \tag{6}$$

This constraint is natural for composite-index construction, including the Serbian indicator tradition in Vukmirovic et al. (2009); Vukmirovic and Ciric (2010); Vukmirovic et al. (2013).

Box-constrained regression:

$$\mathcal{W}_{\text{box}} = [-M, M]^p, \tag{7}$$

for some fixed $M > 0$. This compact parameter space is convenient for the consistency analysis below.

2.3 Loss function specification

For quadratic loss $\ell(y, \hat{y}) = (y - \hat{y})^2$, the expected loss (4) admits an explicit expansion.

Proposition 2.4 (Quadratic loss expansion)**.** *Suppose $\mathbb{E}[\boldsymbol{\xi}_t] = \mathbf{0}$ and $\mathbb{E}[\boldsymbol{\xi}_t \boldsymbol{\xi}_t^\top] = \boldsymbol{\Sigma}_\xi$, with $\boldsymbol{\xi}_t$ independent of $\mathbf{x}_t$. Under quadratic loss,*

$$f(\mathbf{w}) = \frac{1}{T} \sum_{t=1}^{T} (y_t - \mathbf{x}_t^\top \mathbf{w})^2 + \mathbf{w}^\top \boldsymbol{\Sigma}_\xi \mathbf{w}. \tag{8}$$

Proof. Expanding the squared loss yields

$$\begin{aligned}\mathbb{E}_\xi\Big[(y_t - (\mathbf{x}_t + \boldsymbol{\xi}_t)^\top \mathbf{w})^2\Big] &= \mathbb{E}_\xi\Big[(y_t - \mathbf{x}_t^\top \mathbf{w} - \boldsymbol{\xi}_t^\top \mathbf{w})^2\Big] \\ &= (y_t - \mathbf{x}_t^\top \mathbf{w})^2 - 2(y_t - \mathbf{x}_t^\top \mathbf{w})\mathbb{E}[\boldsymbol{\xi}_t^\top \mathbf{w}] + \mathbb{E}[(\boldsymbol{\xi}_t^\top \mathbf{w})^2].\end{aligned}$$

Since $\mathbb{E}[\boldsymbol{\xi}_t] = \mathbf{0}$,

$$\mathbb{E}[\boldsymbol{\xi}_t^\top \mathbf{w}] = 0,$$

and

$$\mathbb{E}[(\boldsymbol{\xi}_t^\top \mathbf{w})^2] = \mathbb{E}[\mathbf{w}^\top \boldsymbol{\xi}_t \boldsymbol{\xi}_t^\top \mathbf{w}] = \mathbf{w}^\top \boldsymbol{\Sigma}_\xi \mathbf{w}.$$

Summing over t and dividing by T gives (8). □

Remark 1. *Proposition 2.4 shows that stochastic optimization under quadratic loss induces a ridge-type regularization term determined by the noise covariance matrix. In the unconstrained case with isotropic noise, $\boldsymbol{\Sigma}_\xi = \sigma^2 \mathbf{I}$, this matches ordinary ridge regression with penalty parameter $\lambda = \sigma^2$. Under simplex or other constraints, the same regularizing structure remains present, but the problem is not identical to unconstrained ridge regression.*

3 Sample Average Approximation

The expected loss (4) generally involves integration over the error distribution. We approximate it by Monte Carlo sampling.

3.1 SAA estimator

Definition 3.1 (Sample Average Approximation)**.** *Let $\{\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(N)}\}$ be i.i.d. draws from the error distribution. The SAA objective is*

$$\hat{f}_N(\mathbf{w}) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{T}\sum_{t=1}^{T} L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t^{(n)}), \tag{9}$$

and the corresponding estimator is

$$\hat{\mathbf{w}}_N = \operatorname*{argmin}_{\mathbf{w}\in\mathcal{W}} \hat{f}_N(\mathbf{w}). \tag{10}$$

3.2 Algorithm

Algorithm 1 Sample Average Approximation for stochastic forecasting

Require: Data $\{(y_t, \mathbf{x}_t)\}_{t=1}^T$, number of Monte Carlo samples N, error distribution P_ξ
Ensure: Estimated weights $\hat{\mathbf{w}}_N$

1: Generate i.i.d. draws $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(N)} \sim P_\xi$
2: Construct perturbed predictors $\tilde{\mathbf{x}}_t^{(n)} = \mathbf{x}_t + \boldsymbol{\xi}_t^{(n)}$ for all t, n
3: Solve

$$\hat{\mathbf{w}}_N = \operatorname*{argmin}_{\mathbf{w}\in\mathcal{W}} \frac{1}{N}\sum_{n=1}^{N}\frac{1}{T}\sum_{t=1}^{T} \ell\big(y_t, (\tilde{\mathbf{x}}_t^{(n)})^\top \mathbf{w}\big)$$

4: **return** $\hat{\mathbf{w}}_N$

4 Assumptions

We now state the regularity conditions used in the consistency analysis.

Assumption 1 (Compact constraint set)**.** *The constraint set $\mathcal{W} \subset \mathbb{R}^p$ is non-empty and compact.*

Assumption 2 (Continuity)**.** *For each realization of $\boldsymbol{\xi}$ and each t, the map $\mathbf{w} \mapsto L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t)$ is continuous on $\mathcal{W}$.*

Assumption 3 (Measurability)**.** *For each $\mathbf{w} \in \mathcal{W}$ and each t, the map $\boldsymbol{\xi}_t \mapsto L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t)$ is measurable.*

Assumption 4 (Integrable envelope)**.** *There exists a measurable function $M(\boldsymbol{\xi})$ such that*

$$|L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t)| \leq M(\boldsymbol{\xi}_t) \quad \text{for all } \mathbf{w} \in \mathcal{W},\ t = 1, \ldots, T,$$

and $\mathbb{E}[M(\boldsymbol{\xi})] < \infty$.

Assumption 5 (Unique optimizer)**.** *The expected loss $f(\mathbf{w})$ has a unique minimizer $\mathbf{w}^* \in \mathcal{W}$.*

Assumption 6 (Noise moments)**.** *The measurement errors $\boldsymbol{\xi}_t$ are i.i.d. with $\mathbb{E}[\boldsymbol{\xi}_t] = \mathbf{0}$ and covariance matrix $\mathbb{E}[\boldsymbol{\xi}_t \boldsymbol{\xi}_t^\top] = \boldsymbol{\Sigma}_\xi$, where $\boldsymbol{\Sigma}_\xi$ is positive semidefinite and has finite eigenvalues.*

4.1 Verification in the GDP forecasting setting

Proposition 4.1 (Assumptions satisfied)**.** *Consider the GDP forecasting problem with:*

- *either the simplex constraint set $\mathcal{W}_{simplex}$ or the box-constrained set $\mathcal{W}_{box}$,*
- *quadratic loss $\ell(y, \hat{y}) = (y - \hat{y})^2$,*
- *Gaussian noise $\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$,*
- *bounded data: $\|\mathbf{x}_t\| \leq B_x$ and $|y_t| \leq B_y$ for all t.*

Then Assumptions 1, 2, 3, 4, and 6 are satisfied. If in addition the expected objective has a unique minimizer on the chosen constraint set, then Assumption 5 also holds.

Proof. For $\mathcal{W}_{\text{simplex}}$, compactness follows because the simplex is closed and bounded. For $\mathcal{W}_{\text{box}} = [-M, M]^p$, compactness is immediate. Continuity holds because quadratic loss is a polynomial in $\mathbf{w}$. Measurability follows from composition of measurable and continuous maps.

For the integrable envelope, note that for $\mathbf{w} \in \mathcal{W}$ there exists a constant $C_W < \infty$ such that $\|\mathbf{w}\| \leq C_W$ for all $\mathbf{w} \in \mathcal{W}$ by compactness. Hence

$$\begin{aligned} |L(\mathbf{w}, \mathbf{x}_t, \boldsymbol{\xi}_t)| &= \Big(y_t - (\mathbf{x}_t + \boldsymbol{\xi}_t)^\top \mathbf{w}\Big)^2 \\ &\leq (|y_t| + \|\mathbf{x}_t + \boldsymbol{\xi}_t\| \, \|\mathbf{w}\|)^2 \\ &\leq (B_y + C_W(B_x + \|\boldsymbol{\xi}_t\|))^2 =: M(\boldsymbol{\xi}_t). \end{aligned}$$

Since Gaussian random vectors have finite second moments, $\mathbb{E}[\|\boldsymbol{\xi}_t\|^2] < \infty$, so $\mathbb{E}[M(\boldsymbol{\xi}_t)] < \infty$. Assumption 6 is immediate from the Gaussian specification. Assumption 5 is a separate identification condition on the expected objective. □

5 Main Convergence Result

5.1 Almost sure convergence

Theorem 5.1 (SAA consistency)**.** *Under Assumptions 1–5, the SAA estimator converges almost surely to the true optimizer:*

$$\hat{\mathbf{w}}_N \xrightarrow{a.s.} \mathbf{w}^* \qquad \text{as } N \to \infty.$$

Proof. The proof proceeds in three steps.

Step 1: Pointwise and uniform convergence of the objective.

By the strong law of large numbers, for each fixed $\mathbf{w} \in \mathcal{W}$,

$$\hat{f}_N(\mathbf{w}) \xrightarrow{a.s.} f(\mathbf{w}).$$

Because $\mathcal{W}$ is compact, the objective is continuous in $\mathbf{w}$, and the family is dominated by an integrable envelope, a uniform law of large numbers applies; see Jennrich (1969). Thus,

$$\sup_{\mathbf{w}\in\mathcal{W}} |\hat{f}_N(\mathbf{w}) - f(\mathbf{w})| \xrightarrow{a.s.} 0. \tag{11}$$

Step 2: Convergence of optimal values.

By (11),

$$\left| \min_{\mathbf{w}\in\mathcal{W}} \hat{f}_N(\mathbf{w}) - \min_{\mathbf{w}\in\mathcal{W}} f(\mathbf{w}) \right| \leq \sup_{\mathbf{w}\in\mathcal{W}} |\hat{f}_N(\mathbf{w}) - f(\mathbf{w})| \xrightarrow{a.s.} 0.$$

Step 3: Convergence of minimizers.

Let $\{\hat{\mathbf{w}}_{N_k}\}$ be any convergent subsequence with limit $\bar{\mathbf{w}}$. Such a subsequence exists by compactness. By definition of $\hat{\mathbf{w}}_{N_k}$,

$$\hat{f}_{N_k}(\hat{\mathbf{w}}_{N_k}) \leq \hat{f}_{N_k}(\mathbf{w}^*).$$

Passing to the limit and using (11) gives

$$f(\bar{\mathbf{w}}) \leq f(\mathbf{w}^*).$$

Since $\mathbf{w}^*$ is the unique minimizer, we must have $\bar{\mathbf{w}} = \mathbf{w}^*$. Every convergent subsequence has the same limit, so the full sequence converges almost surely to $\mathbf{w}^*$. □

Remark 2. *The consistency theorem is the main formal convergence statement in this paper. Stronger finite-sample exponential bounds and sharper optimizer concentration rates generally require additional empirical process assumptions beyond those stated here, especially under Gaussian noise and quadratic loss.*

5.2 Asymptotic normality in the smooth unconstrained case

Theorem 5.2 (Asymptotic normality for smooth interior solutions)**.** *Suppose that:*

1. *the parameter space is open and the optimizer $\mathbf{w}^*$ is an interior point,*
2. *the loss is twice continuously differentiable in $\mathbf{w}$ in a neighborhood of $\mathbf{w}^*$,*
3. *the Hessian $\mathbf{H} = \nabla^2 f(\mathbf{w}^*)$ is nonsingular,*
4. *the score $\nabla L(\mathbf{w}^*, \mathbf{x}_t, \boldsymbol{\xi}_t)$ has finite second moments.*

Then, under standard M-estimation regularity conditions,

$$\sqrt{N}(\hat{\mathbf{w}}_N - \mathbf{w}^*) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \mathbf{H}^{-1}\mathbf{V}\mathbf{H}^{-1}\right), \tag{12}$$

where

$$\mathbf{V} = \mathrm{Var}[\nabla L(\mathbf{w}^*, \mathbf{x}_t, \boldsymbol{\xi}_t)].$$

Proof. This follows from standard M-estimation theory; see van der Vaart (2000). The estimator is defined as the minimizer of an empirical criterion. Under differentiability, identification, and moment conditions, the first-order expansion of the sample score around $\mathbf{w}^*$ combined with a central limit theorem yields the stated asymptotic distribution. □

Remark 3. *The asymptotic normality result is stated only for the smooth unconstrained interior case. Boundary-constrained problems, such as simplex-constrained composite-index estimation, generally require additional arguments and may lead to nonstandard limiting behavior.*

6 Extensions

6.1 Two-stage estimation for composite indices

For composite-index construction, one may use a two-stage procedure:

Stage 1: Estimate simplex-constrained weights

$$\hat{\mathbf{w}}_N = \underset{\mathbf{w}\in\mathcal{W}_{\text{simplex}}}{\operatorname{argmin}} \ \hat{f}_N(\mathbf{w}).$$

Stage 2: Regress the target on the resulting composite indicator

$$y_t = \alpha + \beta C_t + \varepsilon_t, \qquad C_t = \mathbf{x}_t^\top \hat{\mathbf{w}}_N.$$

This formulation is directly connected to the Serbian composite-index literature of Vukmirovic et al. (2009); Vukmirovic and Ciric (2010); Vukmirovic et al. (2013), but replaces deterministic weighting by stochastic estimation under measurement uncertainty.

6.2 Stochastic lasso

We also consider an ℓ_1-regularized stochastic formulation:

$$\hat{\mathbf{w}}_N^\lambda = \underset{\mathbf{w}\in\mathbb{R}^p}{\operatorname{argmin}} \, \hat{f}_N(\mathbf{w}) + \lambda \left\|\mathbf{w}\right\|_1 . \tag{13}$$

This combines sparsity-inducing regularization with robustness to measurement error. In the empirical application, the stochastic lasso specification is the main forecasting model of interest.

Remark 4. *A full consistency proof for stochastic lasso requires assumptions tailored to penalized M-estimation, including conditions on the tuning parameter sequence, identification, and suitable regularity of the penalized objective. Since the empirical section of this paper uses stochastic lasso primarily as a forecasting specification, we do not state a separate proof here.*

7 Numerical Application: Serbian GDP Forecasting

We apply the framework to forecasting Serbian quarterly GDP growth.

7.1 Data description

Data are obtained from the Federal Reserve Economic Database (FRED) for the period 2000Q1–2025Q3.

Table 1: Data sources and variables

Variable	**FRED code**	**Frequency**	**Transformation**
Serbia Real GDP	CLVMNACNSAB1GQRS	Quarterly	% change
Serbia House Prices	QRSR628BIS	Quarterly	% change
Germany Real GDP	CLVMNACSCAB1GQDE	Quarterly	% change
Brent Oil Price	DCOILBRENTEU	Daily	Quarterly average, % change

Table 2: Descriptive statistics (quarterly growth rates)

Variable	**Mean**	**Std**	**Min**	**Max**	**Obs**
Serbia GDP	0.0105	0.0730	−0.177	0.112	99
House Prices	0.0089	0.0412	−0.095	0.143	99
Germany GDP	0.0042	0.0298	−0.113	0.082	99
Oil Price	0.0187	0.1654	−0.541	0.412	99

7.2 Predictor construction

We construct the lagged predictor vector

$$\mathbf{x}_t = \begin{pmatrix} y_{t-1}^{RS} & y_{t-2}^{RS} & y_{t-3}^{RS} & HP_{t-1} & y_{t-1}^{DE} & Oil_{t-1} \end{pmatrix}^\top,$$

where y_t^{RS} is Serbian GDP growth, HP_t is house-price growth, y_t^{DE} is German GDP growth, and Oil_t is oil-price growth.

After constructing lags and removing missing values, we obtain $T = 99$ quarterly observations spanning 2002Q3–2025Q3.

7.3 Experimental setup

Evaluation protocol: rolling-origin out-of-sample evaluation.

- Minimum training window: 40 observations
- Out-of-sample forecasts: 59
- Re-estimation at each step

SAA parameters:

- Number of Monte Carlo samples: $N = 100$
- Noise standard deviation: $\sigma = 0.03$
- Noise distribution: $\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$

Benchmark models:

1. Random Walk: $\hat{y}_t = y_{t-1}$
2. OLS
3. Ridge with fixed penalty
4. Lasso with fixed penalty
5. Stochastic Lasso
6. Ensemble Stochastic
7. Stochastic ElasticNet

7.4 Results

Table 3: Out-of-sample forecast performance (59 rolling forecasts)

Model	**RMSE**	**R^2**	**Dir. Acc. (%)**	**Rank**
Lasso	0.0329	0.718	94.9	1
Stochastic Lasso	**0.0330**	**0.718**	**94.9**	**2**
Ensemble Stochastic	0.0332	0.713	94.9	3
Stochastic ElasticNet	0.0332	0.713	94.9	4
Ridge	0.0335	0.708	94.9	5
OLS	0.0336	0.707	94.9	6
Random Walk	0.1034	−1.779	49.2	7

Table 4: Statistical significance tests (Diebold–Mariano)

Comparison	t**-statistic**	p**-value**	**Conclusion**
Stoch. Lasso vs Lasso	-0.758	0.452	No significant difference
Stoch. Lasso vs OLS	0.341	0.734	No significant difference
Stoch. Lasso vs Ridge	0.287	0.775	No significant difference
Stoch. Lasso vs Random Walk	5.89	< 0.001	Stoch. Lasso better

7.5 Crisis-period analysis

Table 5: Performance by economic period

	RMSE		
Model	**Pre-COVID** (2015–2019)	**COVID** (2020–2021)	**Post-COVID** (2022–2025)
Lasso	0.0192	0.0523	0.0153
Stochastic Lasso	0.0195	0.0518	0.0157
OLS	0.0188	0.0550	0.0143
Random Walk	0.0945	0.0969	0.1030
Observations	20	8	15

7.6 Discussion

The empirical results support four main conclusions.

1. **Statistical parity with strong benchmarks.** Stochastic Lasso achieves RMSE = 0.0330, which is statistically indistinguishable from standard Lasso (RMSE = 0.0329, $p = 0.452$), Ridge, and OLS. This empirical result is consistent with Proposition 2.4, which shows that quadratic-loss stochastic optimization introduces a ridge-type regularization component.

2. **Clear improvement over a naive benchmark.** All regularized and stochastic methods substantially outperform the Random Walk benchmark, with the stochastic lasso producing a large reduction in forecast error relative to the naive alternative.

3. **A narrow predictive spread among well-specified linear models.** In this low-dimensional setting, Serbian GDP lags already contain much of the predictive signal. Once appropriate regularization is used, performance differences among Lasso, Ridge, OLS, and stochastic variants become small.

4. **Only limited evidence on crisis robustness.** During the COVID subsample, Stochastic Lasso slightly improves on standard Lasso (0.0518 vs. 0.0523), but the crisis window contains only 8 observations. This should therefore be interpreted as suggestive rather than conclusive evidence of robustness under structural stress.

A further note concerns directional accuracy. The reported directional accuracy of approximately 94.9% for several top models is high for a macroeconomic forecasting exercise. This likely reflects strong persistence in the sign of quarterly GDP growth during much of the sample, so the directional-accuracy metric should be interpreted alongside RMSE and R^2, not in isolation.

Finally, relative to the earlier Serbian composite-index literature (Vukmirovic et al., 2009; Vukmirovic and Ciric, 2010; Vukmirovic et al., 2013, 2015), the present empirical contribution

is not that a new stochastic method dramatically outperforms all prior approaches. Rather, the contribution is that the traditional Serbian indicator-combination problem is re-cast in a stochastic optimization framework that remains empirically competitive while being conceptually better aligned with measurement uncertainty and revision risk.

8 Conclusion

This paper develops a stochastic optimization framework for macroeconomic forecasting that explicitly accounts for measurement uncertainty in predictor variables. The main formal result is almost sure consistency of the SAA estimator under standard regularity conditions. For quadratic loss, the expected stochastic objective also admits a clear regularization interpretation through the noise covariance matrix.

The empirical application to Serbian GDP forecasting shows that:

- stochastic lasso achieves statistical parity with deterministic benchmark models such as Lasso, Ridge, and OLS,
- stochastic formulations remain clearly superior to a naive random-walk benchmark,
- the main value of the stochastic approach lies less in marginal forecast gains and more in providing a principled framework for estimation under data uncertainty.

Relative to the earlier Serbian literature of Vukmirovic et al. (2009); Vukmirovic and Ciric (2010); Vukmirovic et al. (2013, 2015), this paper should be viewed as a methodological extension rather than a replacement. The composite-index tradition established that Serbian economic activity can be tracked using carefully selected indicator systems. The present paper adds the idea that such indicator-based forecasting problems may be estimated within a stochastic program that explicitly recognizes noisy measurement.

Future work should extend the framework in three directions: first, toward non-Gaussian and time-varying noise structures; second, toward richer real-time nowcasting environments with a larger indicator set; and third, toward formal uncertainty quantification for predictive intervals and revision-sensitive forecasting.

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